

## Use-Case Flow Specification

### *Problem Statement #33 — Micro-Lending & Peer Credit Risk Assessor*

#### **Use Case: Submit Loan Application**

##### **Primary Actor: Borrower**

Secondary Actor: (none — «include»s Compute Credit Risk Score, an internal system use case)

#### **Preconditions**

1. The borrower has a registered and verified account on the platform (KYC completed).
2. The borrower has at least one active financial profile record (income, existing debt, employment status) on file.

#### **Postconditions**

1. A new loan application record exists with a unique application ID and an assigned status (Pending Review, Approved, or Rejected).
2. If approved, a risk tier (Low/Medium/High) and a structured EMI repayment schedule are attached to the application.

#### **Main Success Scenario**

1. Borrower selects "Apply for Loan" from the dashboard.
2. System displays an application form requesting loan amount, tenure, and purpose.
3. Borrower enters the requested details and submits the form.
4. System validates that the borrower's financial profile (income, existing debt, employment status) is complete.
5. System invokes the «include» use case Compute Credit Risk Score, passing debt-to-income ratio, repayment history, and employment stability indicators.
6. System receives the computed risk tier (Low/Medium/High) and assigns a risk-adjusted interest rate.
7. System generates a structured EMI repayment schedule based on principal, interest rate, and tenure.
8. System sets the application status to "Approved" and stores the risk tier and schedule.
9. System notifies the borrower of the approval, assigned risk tier, and repayment schedule.
10. Use case ends successfully.

#### **Alternate Flow — 4a. Incomplete Financial Profile**

- a) System detects one or more missing mandatory financial-profile fields during validation.
- b) System rejects the submission and displays an error message listing the missing fields; no credit approval is generated.
- c) Borrower is redirected to update the financial profile.
- d) Once the profile is completed, flow resumes at step 3 (application resubmission).